

Section B

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)			layers/sheets of hexagons of C atoms bonded together (1) weak forces between layers (allowing layers to move and making it soft) (1)	2			2		
	(b)	(i)		121 kJ mol ⁻¹	1			1		
		(ii)		enthalpy of atomisation for Cl is 121 (1) correctly constructed energy cycle or expression e.g. $\Delta_f H(\text{NaCl}) = \Delta_{\text{at}} H(\text{Na}) + \text{IE}(\text{Na}) + \frac{1}{2} \text{BE}(\text{Cl}_2) + \text{EA}(\text{Cl}) + \Delta_{\text{latt}} H(\text{NaCl})$ (1) $\Delta_{\text{latt}} H(\text{NaCl}) = -771 \text{ kJ mol}^{-1}$ (1)		3		3	2	
		(iii)		student is incorrect must consider entropy of surroundings as well / must consider effects of enthalpy on entropy of surroundings / Gibbs free energy must be considered (must be negative and this includes enthalpy and entropy) (1) award (1) for either of following entropy change for this reaction will be negative as gas is removed entropy change will be negative as entropy of chlorine / gaseous reactant is greater than entropy of product		1	1	2		